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Université Mohamed Khider – Biskra-
Faculté des Sciences et de la Technologie
Département de Tronc Commun

Module : Logique CS
2ième Années ST
Année universitaire : 2019/2020

Travaux dirigés N°2 : Simplification des fonctions Booléennes

(par diagramme de Karnaugh)

Exercice 1 :

1. Simplifiez les fonctions suivantes en utilisant les tableaux de karnaugh :

b	a	X
0	0	1
0	1	1
1	0	1
1	1	0

c	b	a	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

d	c	b	a	Z
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

2. Donnez le logigramme de chaque fonction.

Exercice N°2 :

-Simplifiez les fonctions suivantes en utilisant les tableaux de karnaugh :

$$F1 = a.b.\bar{c} + \bar{a}.\bar{b}.c + a.\bar{b}.\bar{c} + a.\bar{b}.c$$

$$F2 = a.b.c + \bar{a}.b.c + \bar{a}.\bar{b}.c + a.b.\bar{c}$$

$$F3 = \bar{a}.\bar{b}.c.\bar{d} + \bar{a}.b.\bar{c}.d + a.\bar{b}.c.\bar{d} + \bar{a}.\bar{b}.\bar{c}.\bar{d} + a.b.\bar{c}.d + \bar{a}.b.c.d + a.\bar{b}.\bar{c}.\bar{d}$$

$$F4 = \bar{a}.\bar{b}.\bar{c}.\bar{d} + \bar{a}.\bar{b}.\bar{c}.d + \bar{a}.\bar{b}.c.d + \bar{a}.\bar{b}.c.\bar{d} + a.b.\bar{c}.d + a.b.c.d + a.\bar{b}.\bar{c}.d + a.\bar{b}.c.d + a.\bar{b}.c.\bar{d}$$

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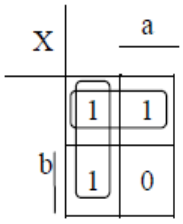
Université Mohamed Khider – Biskra-
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2ième Année ST

Module : Logique CS
Enseignante :Dr Hendaoui Mounira
Année universitaire :2016/2017

Corrigé type des travaux dirigés N°2

Exercice N°1

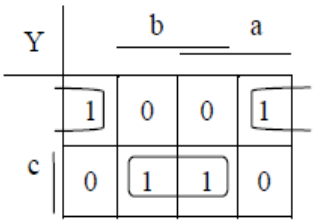
b	a	X
0	0	1
0	1	1
1	0	1
1	1	0



$$X = \bar{a} + \bar{b} = \overline{ab}$$



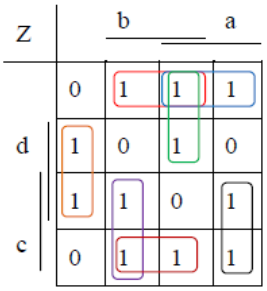
c	b	a	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



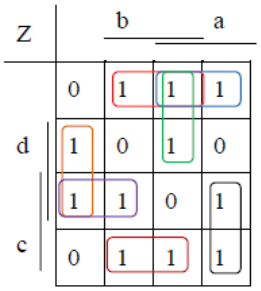
$$Y = bc + \bar{b}\bar{c} = \overline{b \oplus c}$$



d	c	b	a	Z
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



ou

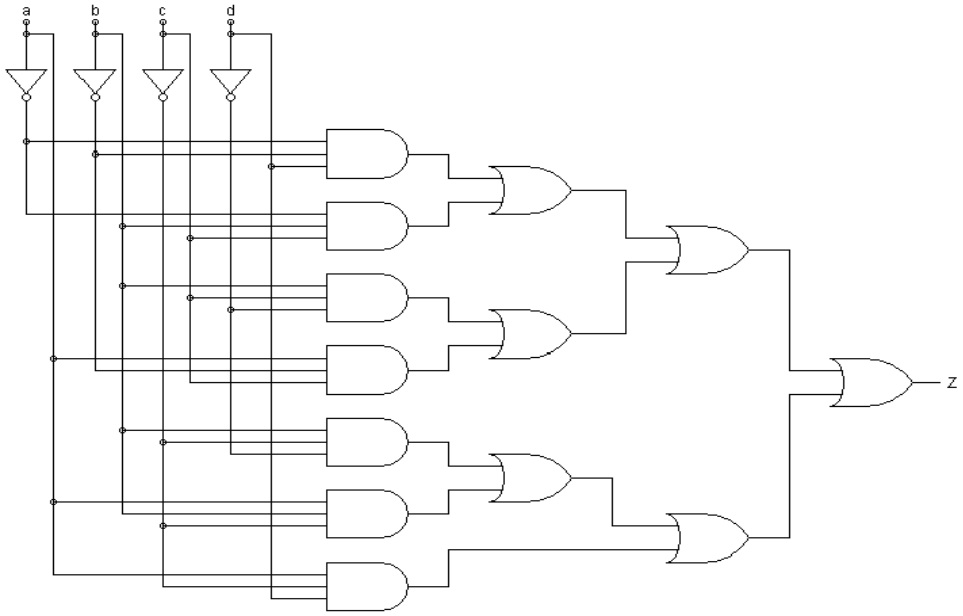


$$Z = \bar{a}\bar{b}d + \bar{a}bc + bc\bar{d} + a\bar{b}c + b\bar{c}\bar{d} + ab\bar{c} + a\bar{c}\bar{d}$$

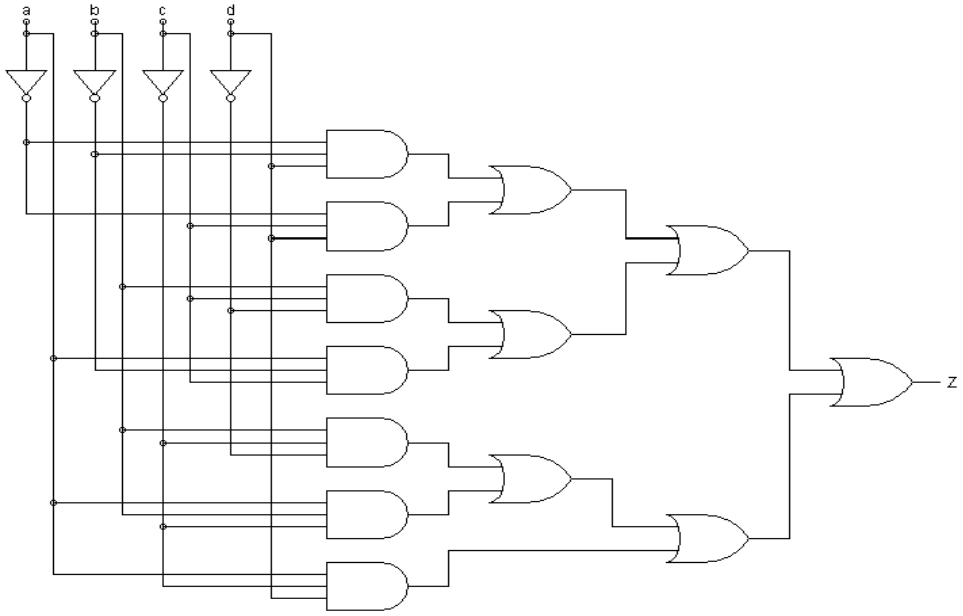
ou

$$Z = \bar{a}\bar{b}d + \bar{a}cd + bc\bar{d} + a\bar{b}c + b\bar{c}\bar{d} + ab\bar{c} + a\bar{c}\bar{d}$$

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ou



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Exercice N°2

$F1 = a.b.\bar{c} + \bar{a}.\bar{b}.c + a.\bar{b}.\bar{c} + a.\bar{b}.c$

F1	b		a	
	0	0	1	1
c	1	0	0	1

$F1 = a\bar{c} + \bar{b}c$

$F2 = a.b.c + \bar{a}.b.c + \bar{a}.\bar{b}.c + a.b.\bar{c}$

F2	b		a	
	0	0	1	0
c	1	1	1	0

$F2 = ab + \bar{a}c$

$F3 = \bar{a}.\bar{b}.c.\bar{d} + \bar{a}.b.\bar{c}.d + a.\bar{b}.c.\bar{d} + \bar{a}.\bar{b}.\bar{c}.\bar{d} + a.b.\bar{c}.d + \bar{a}.b.c.d + a.\bar{b}.\bar{c}.\bar{d}$

F3	b		a	
	1	0	0	1
d	0	1	1	0
	0	1	0	0
c	1	0	0	1

$F3 = \bar{a}bd + b\bar{c}d + \bar{b}\bar{d}$

$F4 = \bar{a}.\bar{b}.\bar{c}.\bar{d} + \bar{a}.\bar{b}.\bar{c}.d + \bar{a}.\bar{b}.c.d + \bar{a}.\bar{b}.c.\bar{d} + a.b.\bar{c}.d + a.b.c.d + a.\bar{b}.\bar{c}.d + a.\bar{b}.c.d + a.\bar{b}.c.\bar{d}$

F4	b		a	
	1	0	0	0
d	1	0	1	1
	1	0	1	1
c	1	0	0	1

$F4 = \bar{a}\bar{b} + ad + a\bar{b}c$

Ou $F4 = \bar{a}\bar{b} + ad + \bar{b}c$.